10. Write a C program for Playfair matrix:

M F H I/J K

U N O P Q

Z V W X Y

E L A R G

D S T B C

Encrypt this message: Must see you over Cadogan West. Coming at once.

Aim:

To encrypt the message “Must see you over Cadogan West. Coming at once.” using the given Playfair matrix**.**

Procedure:

1. Read the plaintext.
2. Remove spaces and punctuation and convert letters to uppercase.
3. Divide the plaintext into pairs.
4. Use the given 5 × 5 matrix to encrypt each pair according to the Playfair rules.
5. Display the ciphertext**.**

Code:

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>
int main() {
   char text[200], out[300];
   char mat[5][5] = {
      {'M','F','H','I','K'},
      {'U','N','O','P','Q'},
      {'Z','V','W','X','Y'},
      {'E','L','A','R','G'},
```

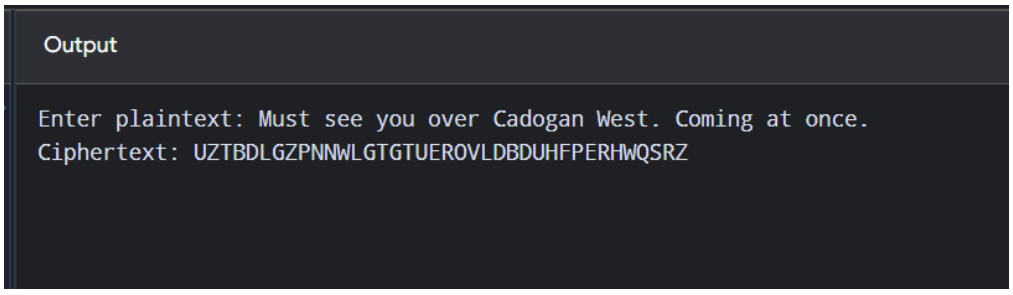
```
    {'D','S','T','B','C'}
};
int r, c, i, n = 0, k = 0;
char p[200];
printf("Enter plaintext: ");
fgets(text, sizeof(text), stdin);
for (i = 0; text[i]; i++)
    if (isalpha((unsigned char)text[i])) {
        p[n] = toupper(text[i]);
        if (p[n] == 'J') p[n] = 'I';
        n++;
    }
p[n] = '\0';
for (i = 0; i < n; i += 2) {
    char a = p[i];
    char b = (i + 1 < n) ? p[i + 1] : 'X';
    int r1 = 0, c1 = 0, r2 = 0, c2 = 0;
    if (a == b) b = 'X';
    for (r = 0; r < 5; r++)
        for (c = 0; c < 5; c++) {
            if (mat[r][c] == a) {
                r1 = r; c1 = c;
            }
            if (mat[r][c] == b) {
                r2 = r; c2 = c;
            }
        }
```

```
        if (r1 == r2) {
            out[k++] = mat[r1][(c1 + 1) % 5];
            out[k++] = mat[r2][(c2 + 1) % 5];
        } else if (c1 == c2) {
            out[k++] = mat[(r1 + 1) % 5][c1];
            out[k++] = mat[(r2 + 1) % 5][c2];
        } else {
            out[k++] = mat[r1][c2];
            out[k++] = mat[r2][c1];
        }
    }
    out[k] = '\0';
    printf("Ciphertext: %s\n", out);
    return 0;
}
```

Input:

Must see you over Cadogan West. Coming at once.

Output:



Result:

Thus, the given Playfair matrix was used to encrypt the plaintext successfully.